

KOSH-AGACH, Russian Federation

WMO: 362590

Lat: 50.0088N

Lon: 88.6721E

Elev: 5776

StdP: 11.88

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.4	-37.4	-46.9	0.4	-41.4	-42.3	0.6	-37.0	17.0	12.4	12.8	1.1	0.6	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	77.2	53.8	74.2	52.8	71.7	52.0	56.7	71.0	55.1	67.9	53.8	66.8	5.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.2	71.9	58.9	50.4	67.2	58.6	48.6	62.8	57.7	26.7	71.0	25.6	68.2	24.7	66.7	64.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	17.6	14.3	DB	-42.6	82.9	5.6	2.8	-46.7	84.9	-50.0	86.5	-53.1	88.1	-57.2	90.1	(4)
(5)			WB	-42.7	59.4	5.6	3.1	-46.7	61.7	-50.0	63.5	-53.1	65.3	-57.2	67.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.7	-17.7	-10.2	13.6	35.2	43.9	57.2	59.7	55.9	44.0	28.9	6.5	-10.7	(6)
(7)		DBStd	28.77	11.46	11.68	13.96	8.47	7.20	6.11	4.81	5.21	7.57	9.63	13.54	10.09	(7)
(8)		HDD50	9646	2100	1686	1130	443	211	13	2	13	206	655	1305	1882	(8)
(9)		HDD65	14361	2565	2106	1595	893	653	240	172	284	631	1120	1755	2347	(9)
(10)		CDD50	775	0	0	0	0	23	230	303	195	24	0	0	0	(10)
(11)		CDD65	14	0	0	0	0	0	6	8	0	0	0	0	0	(11)
(12)		CDH74	276	0	0	0	0	1	104	131	35	5	0	0	0	(12)
(13)		CDH80	21	0	0	0	0	0	10	11	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.0	1.8	2.3	3.5	5.7	6.9	5.2	4.6	4.2	4.1	4.0	3.1	2.2	(14)
(15)	Precipitation	PrecAvg	4.5	0.1	0.1	0.1	0.1	0.3	0.8	1.2	0.9	0.4	0.1	0.2	0.2	(15)
(16)		PrecMax	7.4	0.7	0.3	0.3	0.7	1.5	2.2	2.7	1.9	1.2	0.5	0.9	0.9	(16)
(17)		PrecMin	1.9	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.2	0.2	0.1	0.1	0.2	0.3	0.5	0.6	0.5	0.3	0.1	0.2	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	24.8	47.8	62.2	71.7	81.2	81.3	77.3	73.7	58.5	41.2	21.6	(19)
MCWB			13.5	19.1	34.7	42.9	47.7	55.9	55.1	53.4	49.9	42.0	34.0	18.6	(20)	
2%		DB	10.9	18.7	43.0	57.7	66.0	77.0	77.8	74.6	67.4	53.0	35.5	15.6	(21)	
		MCWB	8.6	14.9	32.2	41.1	45.3	53.1	55.0	53.0	48.0	39.1	30.6	13.5	(22)	
5%		DB	5.4	13.5	38.8	53.4	62.2	73.6	74.2	72.0	62.9	48.2	30.1	9.8	(23)	
		MCWB	3.7	10.8	29.9	38.9	44.1	51.6	53.9	51.9	46.0	37.3	25.7	8.3	(24)	
10%		DB	-1.2	7.9	34.1	49.8	58.9	70.4	71.4	69.1	58.7	43.9	25.2	4.3	(25)	
		MCWB	-2.1	6.0	27.7	36.7	43.1	51.0	52.9	51.2	43.9	34.6	21.7	3.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.6	19.8	36.0	44.2	49.3	58.5	60.5	57.2	51.2	42.9	35.4	19.1	(27)
MDCB			15.9	24.2	45.8	58.8	65.7	76.6	73.9	71.3	67.0	55.6	38.7	20.9	(28)	
2%		WB	8.8	15.0	33.0	42.3	47.3	55.6	57.7	55.1	48.9	40.0	30.7	13.5	(29)	
		MDCB	11.1	18.7	41.9	55.0	62.8	69.9	71.3	67.8	64.0	51.0	35.7	15.4	(30)	
5%		WB	3.6	10.7	30.3	40.0	45.5	54.1	55.7	53.6	47.1	37.6	25.7	8.3	(31)	
		MDCB	5.3	13.2	37.7	51.2	60.4	67.7	68.3	66.9	61.0	47.5	29.7	10.0	(32)	
10%		WB	-2.1	5.9	27.9	37.8	43.5	52.7	54.3	52.4	45.0	34.9	21.8	3.1	(33)	
		MDCB	-1.2	7.9	34.4	48.0	56.6	65.8	66.8	65.5	57.1	43.4	25.1	4.2	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.0	22.7	23.3	21.5	21.4	21.3	20.1	22.2	23.8	21.2	18.3	17.2	(35)
MCDBR			22.3	24.6	24.2	27.6	27.6	26.4	25.2	27.5	30.3	28.6	20.8	20.2	(36)	
MCWBR			20.4	21.3	15.2	13.6	12.5	8.6	8.5	9.9	13.7	16.3	15.6	18.4	(37)	
5% WB		MCDBR	22.4	24.2	22.4	24.8	24.0	22.3	21.6	23.2	27.4	27.0	20.0	20.6	(38)	
		MCWBR	20.6	21.1	14.8	12.8	11.7	7.9	8.4	9.0	12.6	15.9	15.3	18.9	(39)	
		Clear-Sky Solar Irradiance	taub	0.228	0.253	0.301	0.348	0.341	0.341	0.340	0.324	0.289	0.277	0.251	0.230	(40)
taud	2.312		2.265	2.197	2.213	2.299	2.391	2.412	2.429	2.480	2.434	2.354	2.295	(41)		
Ebn at Noon	268		286	286	283	290	291	289	289	291	274	255	248	(42)		
Edh at Noon	21		28	37	41	39	36	35	33	28	24	20	18	(43)		
(44)	All-Sky Solar Radiation	RadAvg	470	797	1266	1703	1994	1985	1901	1694	1320	820	477	361	(44)	
RadStd		22	30	59	74	106	109	88	82	93	70	28	22	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	-2.57	-2.93	N/A	N/A	N/A		

Nomenclature: See separate page